FN6 Worked Examples: Inverse Functions

Recording companion for the FN6 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What does f do to x, and how do I undo it? The inverse undoes the function: if f(a) = b, then $f^{-1}(b) = a$. To find it, write y = f(x), swap x and y, and solve for y. For a power or root, that means undoing the steps in reverse order. For a fraction, clear the denominator, gather the y terms on one side, factor out y, and divide. Check by putting an output of f into your answer: you should get the input back.

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Example 1: A Fifth Power

$$f(x)=(x+3)^5-2$$

Question

What is the inverse of this function?

- **A.** $f^{-1}(x)=(x+2)^{1/5}-3$
- **B.** $f^{-1}(x)=(x-2)^{1/5}+3$
- **C.** $f^{-1}(x)=(x+3)^{1/5}-2$
- **D.** $f^{-1}(x)=1/((x+3)^5-2)$

Answer and reasoning

f adds 3, raises to the 5th power, then subtracts 2. Undo in reverse order: add 2, take the fifth root (the $1/5$ power), then subtract 3. So $f^{-1}(x)=(x+2)^{1/5}-3$. Check: $f(-2)=1^5-2=-1$, and $f^{-1}(-1)=1^{1/5}-3=-2$. $(x-2)^{1/5}+3$ undoes each step with the wrong sign, repeating them instead of reversing them; $(x+3)^{1/5}-2$ keeps the numbers in their original places, undoing the steps in the original order instead of reverse order; and $1/f(x)$ is the reciprocal, which the -1 in f^{-1} does not mean. **$f^{-1}(x)=(x+2)^{1/5}-3$.**

Example 2: A Cube Root

$$g(x)=7+(x-1)^{1/3}$$

Question

What is the inverse of this function?

- **A.** $g^{-1}(x)=(x-7)^3-1$
- **B.** $g^{-1}(x)=(x-7)^3+1$
- **C.** $g^{-1}(x)=(x+7)^3+1$
- **D.** $g^{-1}(x)=(x+7)^3-1$

Answer and reasoning

The $1/3$ power is the cube root. g subtracts 1, takes the cube root, then adds 7. Undo in reverse: subtract 7, cube, then add 1: $g^{-1}(x)=(x-7)^3+1$. Check: $g(28)=7+27^{1/3}=10$, and $g^{-1}(10)=3^3+1=28$. $(x-7)^3-1$ undoes the first step with the wrong sign; $(x+7)^3+1$ undoes the last step with the wrong sign; and $(x+7)^3-1$ gets both signs wrong. **$g^{-1}(x)=(x-7)^3+1$.**

Example 3: The First Step

$$h(x)=(x-5)^7+3$$

Question

To find the inverse, you undo the steps of h. Which step do you undo first?

- **A.** Subtract 3
- **B.** Take the seventh root
- **C.** Add 5
- **D.** Subtract 5

Answer and reasoning

h subtracts 5, raises to the 7th power, then adds 3. The last step done is undone first, like taking off shoes before socks. So subtract 3 first, then take the seventh root, then add 5: $h^{-1}(x)=(x-3)^{1/7}+5$. Taking the seventh root is the second step to undo, not the first; adding 5 undoes the first step of *h*, so it comes last; and subtracting 5 repeats that step instead of undoing it. **Subtract 3.**

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Example 4: A Fifth Root

$$f(x)=(x-3)^{1/5}+2$$

Question

CheckIt writes roots as powers. What is the inverse of this function?

A. $f^{-1}(x)=(x+2)^5-3$
B. $f^{-1}(x)=(x-2)^{1/5}+3$
C. $f^{-1}(x)=(x-2)^5+3$
D. $f^{-1}(x)=(x-3)^5+2$

Answer and reasoning

The $1/5$ power is the fifth root, and raising to the 5th power undoes it. f subtracts 3, takes the fifth root, then adds 2. Undo in reverse: subtract 2, raise to the 5th power, then add 3: $f^{-1}(x)=(x-2)^5+3$. Check: $f(35)=32^{1/5}+2=4$, and $f^{-1}(4)=2^5+3=35$. $(x+2)^5-3$ repeats each addition or subtraction instead of undoing it; $(x-2)^{1/5}+3$ keeps the fifth root, but a root is undone by a power; and $(x-3)^5+2$ undoes the steps in the original order. $\boldsymbol{f^{-1}(x)=(x-2)^5+3}$**.**

Example 5: Fraction 1

$$f(x)=(2x+3)/(x-5)$$

Question

What is the inverse of this function?

- **A.** $f^{-1}(x)=(x-5)/(2x+3)$
- **B.** $f^{-1}(x)=(-5x+3)/(x-2)$
- **C.** $f^{-1}(x)=(5x+3)/(x-2)$
- **D.** $f^{-1}(x)=(5x+3)/(x+2)$

Answer and reasoning

Swap: $x=(2y+3)/(y-5)$. Clear the denominator: $xy-5x=2y+3$. Gather the y terms: $xy-2y=5x+3$. Factor: $y(x-2)=5x+3$, so $y=(5x+3)/(x-2)$. Check: $f(6)=15/1=15$, and $(5(15)+3)/(15-2)=78/13=6$. $(x-5)/(2x+3)$ is $1/f(x)$, the reciprocal; $-5x+3$ moves $-5x$ across without changing its sign; and $x+2$ moves $2y$ across without changing its sign. **$f^{-1}(x)=(5x+3)/(x-2)$.**

Example 6: The Key Move

$$x=(3y-1)/(2y+7)$$
$$2xy+7x=3y-1$$

Question

A student is finding the inverse of f(x) = (3x − 1)/(2x + 7) and has cleared the denominator. What is the next move?

A. Get both y terms on one side: 2xy − 3y = −7x − 1
B. Divide both sides by 2x: y + 7 = (3y − 1)/(2x)
C. Subtract 7x, then divide by 2x: y = (3y − 1 − 7x)/(2x)
D. Cancel the y terms: 2x + 7x = 3 − 1

Answer and reasoning

y appears on both sides, so gather the *y* terms on one side and everything else on the other: $2xy-3y=-7x-1$. Then factor out *y*: $y(2x-3)=-7x-1$, and $f^{-1}(x)=(-7x-1)/(2x-3)$. Dividing by $2x$ does not divide the $7x$ correctly and still leaves *y* on both sides; subtracting $7x$ and dividing gives a "solution" with *y* on the right, so *y* is not solved for; and you cannot cancel *y* from terms that are added or subtracted. **Get both *y* terms on one side.**

Example 7: Fraction 2

$$g(x)=(-x+5)/(3x+2)$$

Question

What is the inverse of this function?

- **A.** $g^{-1}(x)=(2x+5)/(3x+1)$
- **B.** $g^{-1}(x)=(-2x+5)/(3x+1)$
- **C.** $g^{-1}(x)=(-2x+5)/(3x-1)$
- **D.** $g^{-1}(x)=(3x+2)/(-x+5)$

Answer and reasoning

Swap: $x=(-y+5)/(3y+2)$. Clear: $3xy+2x=-y+5$. Gather: $3xy+y=-2x+5$. Factor: $y(3x+1)=-2x+5$, so $g^{-1}(x)=(-2x+5)/(3x+1)$. CheckIt may print this as $-(2x-5)/(3x+1)$, which is the same. $2x+5$ moves $2x$ across without changing its sign; $3x-1$ moves $-y$ across without changing its sign; and $(3x+2)/(-x+5)$ is the reciprocal.
***$g^{-1}(x)=(-2x+5)/(3x+1)$*.**

Example 8: Check with a Point

$$f(x)=(x+7)/(5x-3), \quad f(2)=9/7$$

Question

Since f(2) = 9/7, the inverse must send 9/7 back to 2. Which candidate does that?

- **A.** $(5x-3)/(x+7)$
- **B.** $(3x-7)/(5x-1)$
- **C.** $(3x+7)/(5x-1)$
- **D.** $(-3x+7)/(5x-1)$

Answer and reasoning

Put in $x=9/7$. For $(3x+7)/(5x-1)$: the top is $27/7+49/7=76/7$ and the bottom is $45/7-7/7=38/7$, so it gives $76/38=2$. That is the inverse, and solving confirms it: $5xy-3x=y+7$, $y(5x-1)=3x+7$. The reciprocal $(5x-3)/(x+7)$ gives $(24/7)/(58/7)=12/29$; $(3x-7)/(5x-1)$, which moves the 7 with the wrong sign, gives $-11/19$; and $(-3x+7)/(5x-1)$, which moves $3x$ with the wrong sign, gives $11/19$. ***(3x+7)/(5x−1).***

Example 9: The Notation

$$f^{-1}(x)$$

Question

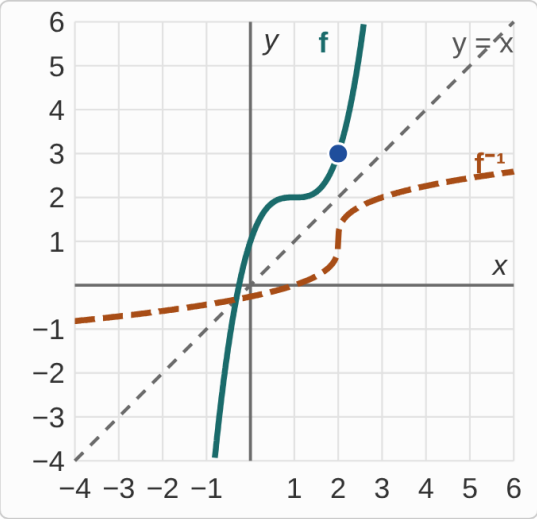
Which statement about f^{-1} is true?

- **A.** $f^{-1}(x)$ means $1/f(x)$, the reciprocal of the function f
- **B.** f^{-1} undoes f: if $f(2) = 9$, then $f^{-1}(9) = 2$
- **C.** $f^{-1}(x)$ means $-f(x)$, the opposite of f
- **D.** f^{-1} is f with every x replaced by $-x$

Answer and reasoning

The -1 in f^{-1} is a name, not an exponent: f^{-1} is the function that undoes f. If f sends 2 to 9, f^{-1} sends 9 back to 2. The reciprocal $1/f(x)$ is a different function, written $(f(x))^{-1}$; $-f(x)$ flips the graph over the x-axis; and replacing every x by $-x$ gives $f(-x)$, which flips it over the y-axis. The graph of f^{-1} is the flip of f over the line $y=x$. **f^{-1} undoes f.**

Example 10: A Point on Each Graph



Plotted data: A grid with x and y from −4 to 6. A solid curve labeled f rises from lower left to upper right through (0, 1), (1, 2) and (2, 3), flattening out near (1, 2); the point (2, 3) on it is marked with a dot. A long-dashed curve labeled f^{-1} rises through (1, 0), (2, 1) and (3, 2), standing nearly upright near (2, 1). A short-dashed line labeled y = x runs between them, and each curve is the mirror image of the other across it.

$$f(x)=(x-1)^3+2$$

Question

The point (2, 3) is on the graph of f. Which point must be on the graph of f^{-1}?

- **A.** *(3,2)*
- **B.** *(2,3)*
- **C.** *(−2,−3)*
- **D.** *(2,1/3)*

Answer and reasoning

$f(2)=1^3+2=3$, so f sends 2 to 3, and f^{-1} sends 3 back to 2: the point $(3,2)$. Swapping the coordinates is the mirror image across $y=x$. Check with the inverse, $f^{-1}(x)=(x-2)^{1/3}+1$: $f^{-1}(3)=1+1=2$. $(2,3)$ is on f, and on f^{-1} only if the two graphs cross there, which they do

not; *(−2,−3)* changes the signs instead of swapping; and *(2,1/3)* takes the reciprocal of the output. ***(3,2)*.**

Example 11: Why Swap

$$y=f(x) \longrightarrow x=f(y)$$

Question

When you find an inverse, why do you swap x and y?

A. The inverse takes the outputs of f as its inputs
B. Swapping makes the equation easier to solve
C. The inverse is the flip of f over the x-axis
D. The inverse is the reciprocal of f, so x and y trade places

Answer and reasoning

f turns inputs x into outputs y. The inverse runs the other way: its inputs are the outputs of f, and its outputs are the inputs of f. Swapping the letters writes that down, and solving for the new y gives a formula for $f^{-1}(x)$. Swapping does not make the algebra easier, it changes which question you are answering; the flip over the x-axis is $-f(x)$, but swapping coordinates flips over the line $y=x$; and the inverse is not the reciprocal.
The inverse takes the outputs of *f* as its inputs.

Example 12: One-to-One

$$f(x)=a+(x+b)^r$$

Question

On CheckIt, r is always 3, 5, 7, 1/3, 1/5 or 1/7. Why does every such function have an inverse?

A. Every function has an inverse
B. These functions have no fractions in them
C. The exponent r is always a positive number
D. Odd powers and roots never repeat outputs

Answer and reasoning

A function has an inverse when it is one-to-one: no two inputs give the same output, so each output leads back to exactly one input. Odd powers and odd roots keep the sign: $2^3=8$ but $(-2)^3=-8$, so they never repeat an output. Not every function has an inverse, though: x^2 gives 4 for both 2 and -2, so 4 cannot be sent back to one input; fractions are fine, since the fractions in the other kind of CheckIt problem are one-to-one too; and x^2 also has a positive exponent, so the exponent being a positive number is not the reason. **Odd powers and odd roots never repeat an output.**

Example 13: Slip 1

$$x=(3y+2)/(y-7)$$
$$xy-7x=3y+2$$
$$xy-3y=2-7x$$
$$y=(2-7x)/(x-3)$$

Question

A student found the inverse of f(x) = (3x + 2)/(x − 7). What is the mistake, if any?

A. There is no mistake.
B. Moving −7x across gives +7x: (7x + 2)/(x − 3).
C. The inverse is the reciprocal of f, (x − 7)/(3x + 2).
D. Cancel the y terms instead, which gives x − 7x = 3 + 2.

Answer and reasoning

From $xy-7x=3y+2$, add $7x$ to both sides and subtract $3y$: $xy-3y=7x+2$. The student moved $-7x$ but kept its minus sign. So $y(x-3)=7x+2$ and $f^{-1}(x)=(7x+2)/(x-3)$. Check: $f(8)=26$, and $(7(26)+2)/(26-3)=184/23=8$. The reciprocal is $1/f(x)$, not the inverse; and y terms that are added or subtracted cannot be canceled. ***$f^{-1}(x)=(7x+2)/(x-3)$*.**

Example 14: Slip 2

$$f(x)=(x-5)^3+7$$
$$\text{undo } -5\text{: add } 5$$
$$\text{undo the cube: cube root}$$
$$\text{undo } +7\text{: subtract } 7$$
$$f^{-1}(x)=(x+5)^{1/3}-7$$

Question

A student found this inverse by undoing the steps. What is the mistake, if any?

A. There is no mistake.
B. A cube is undone by a square root, not a cube root.
C. Undo in reverse order: f⁻¹(x) = (x − 7)^(1/3) + 5.
D. The signs are wrong: f⁻¹(x) = (x − 5)^(1/3) + 7.

Answer and reasoning

Each undo step is right, but they are in the wrong order. *f* subtracts 5, cubes, then adds 7, so the inverse undoes the last step first: subtract 7, take the cube root, then add 5: $f^{-1}(x)=(x-7)^{1/3}+5$. Check: $f(6)=1+7=8$, and $(8-7)^{1/3}+5=6$, but the student's version gives $13^{1/3}-7$, not 6. A cube root undoes a cube, not a square root; and $(x-5)^{1/3}+7$ repeats each step of *f* instead of undoing it. **$f^{-1}(x)=(x-7)^{1/3}+5$.**

Example 15: Slip 3

$$x=2+(y+5)^{1/7}$$
$$x-2=(y+5)^{1/7}$$
$$(x-2)^7=y+5$$
$$y=(x-2)^7-5$$

Question

A student found the inverse of h(x) = 2 + (x + 5)^(1/7) by swapping and solving. What is the mistake, if any?

A. There is no mistake.
B. Raising to the 7th power should give (x − 2)^(1/7).
C. The 2 should be subtracted last: y = (x − 5)^7 − 2.
D. The inverse is 1/h(x), so swapping is not needed.

Answer and reasoning

Swap, subtract 2, raise both sides to the 7th power (which undoes the *1/7* power), and subtract 5: $h^{-1}(x)=(x-2)^7-5$. Check: $h(-4)=2+1=3$, and $(3-2)^7-5=-4$. The student is right. The 7th power undoes the *1/7* power, so the root is gone, not kept; the 2 was added last in *h*, so it is undone first, as the student did; and the inverse is not the reciprocal. **There is no mistake.**

Example 16: Slip 4

$$f(x)=(5x-1)/(2x+3)$$
$$f^{-1}(x)=(2x+3)/(5x-1)$$

Question

A student wrote this inverse. What is the mistake, if any?

A. There is no mistake.
B. It should be the opposite of f instead: −(5x − 1)/(2x + 3).
C. Only the x terms trade places: (2x − 1)/(5x + 3).
D. That is 1/f(x); solving gives (−3x − 1)/(2x − 5).

Answer and reasoning

Flipping the fraction gives $1/f(x)$, the reciprocal, not the inverse. Swap and solve: $x(2y+3)=5y-1$, so $2xy+3x=5y-1$, $2xy-5y=-3x-1$, $y(2x-5)=-3x-1$, and $f^{-1}(x)=(-3x-1)/(2x-5)$. Check: $f(1)=4/5$, and $(-12/5-1)/(8/5-5)=(-17/5)/(-17/5)=1$; the student's version gives $(23/5)/3=23/15$. $-f(x)$ is the opposite, also not the inverse; and swapping only the x terms is not a rule for anything. ***$f^{-1}(x)=(-3x-1)/(2x-5)$*.**

Example 17: Slip 5

$$y(5x+1)=2-7x$$
$$y=(2-7x)/(5x+1)=(2-7)/(5+1)=-5/6$$

Question

A student was finding the inverse of f(x) = (−x + 2)/(5x + 7) and got this far. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** The x's cannot be canceled: $f^{-1}(x) = (2 - 7x)/(5x + 1)$.
- **C.** The y-term should be y(5x − 1), not y(5x + 1).
- **D.** An inverse is a number, so −5/6 is right but should be 5/6.

Answer and reasoning

Everything up to $y=(2-7x)/(5x+1)$ is right: $5xy+7x=-y+2$, so $5xy+y=2-7x$. The last step is the slip. The x in $7x$ and in $5x$ are parts of sums, not factors of the whole top and bottom, so they cannot be canceled. The inverse is a function of x: $f^{-1}(x)=(2-7x)/(5x+1)$. Moving $-y$ across gives $+y$, so the factor is $5x+1$; and an inverse function is a formula, not a single number. ***$f^{-1}(x)=(2-7x)/(5x+1)$*.**